Santiago Osorio Jurado

Neural Engineering · Brain-Computer Interfaces · Neuromodulation
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RESEARCH INTERESTS

Neural interfaces and neuromodulation: signal processing and machine learning to recover real-time sensory and visual feedback from sparse, low-SNR neural signals; subject-specific electromagnetic field modeling for more precise stimulation; and implantable and wearable devices for neurorehabilitation, built to restore lost function rather than only manage symptoms.

EDUCATION

Bachelor of Engineering (B.E.), Biomedical Engineering Aug. 2022 – May 2026
The City College of New York, CUNY | Grove School of Engineering New York, NY

- Magna Cum Laude; Dean's List; Tau Beta Pi Engineering Honor Society (top 12.5% of class, 2025).
- Relevant coursework: Neural Engineering and Applied Bioelectricity; Imaging and Signal Processing; Biomedical Transducers and Instrumentation; Bioelectrical Circuits; Dynamical Systems and Modeling.

RESEARCH EXPERIENCE

Visiting Researcher May 2026 – Present
KTH Royal Institute of Technology | PI: Prof. Rodrigo Moreno Stockholm, Sweden

- Built a subject-specific finite-element pipeline (SimNIBS) deriving anisotropic white- and grey-matter conductivity tensors from b-tensor-encoded diffusion MRI (QTI); first reported use of multidimensional diffusion MRI (MD-dMRI) conductivity in a human tDCS field model.
- Compared three conductivity formulations (isotropic, single-shell DTI, QTI mean tensor) across four electrode montages in a 29-participant cohort (12 Parkinson's disease, 17 controls); built the workflow in Python, MATLAB, and Bash over FreeSurfer and FSL.
- Cross-validated field predictions against MR elastography; first-author manuscript in preparation.

Neuroengineering Researcher Jan. 2025 – May 2026
CCNY Neural Engineering Group | PI: Prof. Marom Bikson New York, NY

- Engineered multilayer Zn/AgCl redox electrodes with a hydrogel interface ($\leq 600\ \Omega\cdot$cm) enabling safe stimulation at 6 mA, triple the conventional 2 mA dose; verified electrochemical stability across 1–10 mA via EIS, overpotential sweeps, and SEM.
- Ran IRB-approved single-blind randomized crossover sessions (n = 5; 1–6 mA), collecting VAS pain, Draize erythema, thermal imaging, and PANAS measures to produce first-in-cohort tolerability data.
- Developed the MRI-derived COMSOL field models (1.95–3.25 V/m cortical fields at 6 mA) and the custom PCB stimulator hardware; quantified dose-response with linear mixed-effects models in Python. Co-author on the resulting bioRxiv preprint.

Machine Learning Researcher (SUPERB Fellowship) Jun. – Dec. 2025
Computational Imaging Lab, UC Berkeley | PI: Prof. Laura Waller Berkeley, CA

- Benchmarked a U-Net against a CNN-transformer hybrid in PyTorch for denoising low-SNR darkfield microscopy; identified the U-Net as the stronger model by PSNR, SSIM, and error maps on simulated and experimental targets.
- Trained both models with no clean ground-truth images available, by generating a 768-pair synthetic dataset (gamma correction and Poisson-Gaussian noise) and packaging training in a reproducible, end-to-end pipeline.

Medical Imaging Researcher (NSF IRES) May – Sep. 2024
KTH Royal Institute of Technology | PI: Prof. Mats Persson Stockholm, Sweden

- Identified 90 kVp with 40 keV virtual monoenergetic reconstruction as optimal for iodine contrast-to-noise on a Siemens NAEOTOM Alpha photon-counting CT (70–140 kVp, two dose levels, multi-energy phantoms at 0.5–20 mg/mL).
- Built an automated Python workflow comparing QIR0/QIR4 reconstruction modes and authored a physics-based parameter-selection guide for radiologists.

PUBLICATIONS & PREPRINTS

Donnery, K., FallahRad, M., Khadka, N., Saw, M., Babaev, B., **Osorio, S.**, Belali Koochesfahani, M., Bhuiyan, R., Elwassif, J. M., & Bikson, M. (2025). High-capacity transcranial direct current stimulation (HC-tDCS). *bioRxiv*. https://doi.org/10.1101/2025.06.11.659142

Osorio Jurado, S., Olsson, C., & Moreno, R. (in preparation). Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI. Target: *NeuroImage: Clinical.*

CONFERENCE PRESENTATIONS & POSTERS

Osorio, S., et al. (2025, November). High-capacity transcranial direct current stimulation (HC-tDCS) [Poster]. Annual Biomedical Research Conference for Minoritized Scientists (ABRCMS), San Antonio, TX. (NIH U-RISE)

Osorio, S. (2025, August). Deep-learning denoising for extreme short-exposure darkfield microscopy [Poster and oral presentation]. SUPERB Summer Research Symposium, UC Berkeley, Berkeley, CA.

Osorio, S. (2024). Optimal tube-voltage selection in photon-counting CT for iodine contrast imaging [Poster]. NSF IRES Research Symposium, KTH / CCNY.

SELECTED PROJECTS

Thermal Stimulator for Automated Rodent Pain Testing (Senior Design Capstone) Aug. 2025 – May 2026
CCNY Biomedical Engineering | Sponsor: Dr. Justin Burdge, Tactorum New York, NY

- Enabled precise, repeatable heat and cold stimuli for automated rodent pain testing, holding temperature to within ±0.5 °C and tracking a 1 °C-per-second ramp, by writing the control software that reads a temperature sensor and drives the heater through a reconfigurable four-stage cycle (cool, preheat, ramp, hold) adjustable over a serial connection while it runs.
- Restored accurate ramp tracking after the temperature consistently fell short, by tracing the fault to a control-logic reset at the constant-to-ramp handoff and retuning the feedback (PID) controller.
- Built and validated the Arduino-based power electronics that drive the heater (checked against a reference thermometer) and delivered them as a plug-in module for the sponsor's rodent-testing platform (Tactorum ARM); with a thermoelectric (Peltier) cooling stage the team added, between-test cool-down fell from 417 to 66 second.

TEACHING & MENTORSHIP

Undergraduate Research Mentor, BMES ResearchConnect 2024 – 2026
The City College of New York New York, NY

- Mentored and placed five undergraduates, many first-generation, into faculty research positions through the ResearchConnect student-faculty matching program.

Biology Laboratory Assistant Aug. 2023 – May 2024
The City College of New York New York, NY

- Supported lab courses for 100+ undergraduates per semester; handled instrument calibration and BSL-2 reagent preparation.

LEADERSHIP & SERVICE

Chapter Officer (Secretary), Biomedical Engineering Society (BMES) 2024 – 2026
CCNY Chapter New York, NY

- Co-organized the Biodesign Challenge, a medical-device prototyping competition (56 students; $5,500 in awards) addressing real-world clinical needs.

Chapter Officer (Secretary), Tau Beta Pi Engineering Honor Society 2025 – 2026
CCNY Chapter (New York Eta) New York, NY

- Represented the chapter at the Tau Beta Pi District 2 Conference and National Convention; supported member initiation and chapter operations.

AWARDS, GRANTS & SCHOLARSHIPS

Rukin Award for Academic and Professional Perseverance in BME, CCNY	2026
BMES Student Chapter Executive Board Award, CCNY	2026
Wallace H. Coulter Award for Academic Service in BME, CCNY	2025, 2026
SUPERB Summer Research Fellowship, UC Berkeley	2025
NIH U-RISE Scholar (NIGMS T34), National Institutes of Health	2025
NSF IRES Fellowship, KTH Royal Institute of Technology / CCNY	2024

RELEVANT SKILLS

Programming & computing: Python (NumPy, SciPy, pandas, PyTorch, Matplotlib), MATLAB, C/C++ (Arduino), Bash; Git; LaTeX; HPC / SLURM clusters.
Neural & biomedical modeling: finite-element field modeling (COMSOL, SimNIBS); MRI processing (FreeSurfer, FSL, ANTs, SAMSEG); diffusion MRI (DTI, QTI / b-tensor encoding); anisotropic conductivity estimation; statistical modeling (linear mixed-effects, power and FDR analysis).
Machine learning & signal processing: deep learning (U-Net, CNN-transformer) in PyTorch; image denoising and reconstruction; synthetic data generation; PSNR/SSIM evaluation.
Electronics & devices: PCB design and stimulator electronics; electrode fabrication; electrochemical impedance spectroscopy (EIS); SEM; embedded systems (Arduino, K-type thermocouples, MOSFET drivers, SPI); closed-loop PID/PWM control; 3D CAD (SolidWorks) and printing.
Experimental & translational: IRB human-subjects protocols; single-blind randomized crossover design; tolerability and sensory assays (VAS, Draize, PANAS); photon-counting CT acquisition and analysis.
Languages: Spanish (native); English (fluent, professional).